

MAIC - Introduction to Python - Quick Reference

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Variables

Syntax	Examples
<variable name> = <value>	first_name = 'Jude'
	last_name = 'Bellingham'
	age = 20

Print Function

Syntax	Examples
print(<value>)	print('Hello world')
	print('My name is Jude')
	print(2 + 8)
print(<variable name>)	print(first_name)
	print(age)

Arithmetic Operators

Arithmetic	Operator	Examples
Addition	+	a = 2 + 8
Subtraction	-	b = 10 - 6
Multiplication	*	c = 5 * 4
Division	/	d = 12 / 3

Comparison Operators

Comparison	Operator	Examples
Equal to	==	print(10 == 100)
Not equal to	!=	print(15 != 20)
Greater than	>	print(2 > 10)
Less than	<	print(10 < 100)
Greater than or equal to	>=	print(10 >= 10)
Less than or equal to	<=	print(20 <= 19)

Functions

Syntax	Examples
<pre>def <function name>(<inputs>): <function body> return <output></pre>	<pre>def add(x, y): sum = x + y return sum add(10, 18)</pre>
	<pre>def convert_cm_to_ft(cm): ft = cm / 30.48 return ft height_cm = 200 height_ft = convert_cm_to_ft(height_cm) print(height_ft)</pre>

Conditional Statements

Syntax	Examples
<pre>if <condition>: <true outcome> else: <false outcome></pre>	<pre>age = int(input('Please enter your age: ')) if age >= 18: print('You are allowed to vote.') ... else: print('You are not allowed to vote.')</pre>
<pre>if <condition>: <true outcome> elif <condition>: <true outcome> else: <false outcome></pre>	<pre>score = int(input('Please enter your score: ')) if score >= 90: grade = 'A' elif score >= 80 and score < 90: grade = 'B' elif score >= 70 and score < 80: grade = 'C' else: grade = 'D'</pre>

Data Structures

Data Structure	Examples
List	<pre>shopping_list = ['apples', 'bananas', 'milk'] print(shopping_list[0]) print(shopping_list[0:2]) print(len(shopping_list)) if 'bananas' in shopping_list: print('bananas are in my shopping list')</pre>

Loops

Loop	Examples
For Loop	<pre>for item in shopping_list: print(item) ...</pre>